Chapter 2

Computational Foundation

2.1 Scope and Review Axes

Chapter 1 described the circuit as a whole: the laminar column, its excitatory and inhibitory cell classes, and the cortico-thalamo-cortical loop. This chapter turns to the computational side and asks how existing models represent cortical inhibition, sparse coding and associative memory, and what they have accomplished. Rather than cataloguing these models, it reads each one for what it establishes and also for where it stops, so that the boundary of current work is visible when the gap is stated.

The review is organised along three axes, the first of which is the *substrate*: the medium in which a model represents and computes. Candidates include binary sparse distributed representations over minicolumns with dendritic segments, rate-coded populations, spiking attractor networks, and multi-compartment neurons. The substrate matters because a mechanism shown in one medium does not automatically carry over to another. The second axis is the *inhibition model*. Inhibition is either homogeneous, a single mechanism such as k-winners-take-all applied uniformly, or differentiated, with distinct interneuron classes that differ in target compartment, dynamics and arithmetic effect. The third axis is *encoding versus retrieval*. It asks how a model relates the storage of new associations to the recall of stored ones, and what, if anything, decides between them.

Because the thesis turns on these three terms, they are defined here. *Encoding* is the process by which a new input pattern is turned into a stable representation and is stored, here through synaptic change. *Retrieval* is the process by which a partial or noisy cue reactivates a stored representation. The two are not mutually exclusive: a network can retrieve while its synapses remain plastic, and encoding a new input usually involves matching it against what is already stored. *Mode arbitration* is the mechanism that decides which of the two dominates at a given moment, for example by gating plasticity or by weighting external input against recurrent or feedback input. The third

axis therefore asks less whether a model encodes and retrieves than whether it arbitrates between the two, and by what means.

The third axis has a clear precedent outside the neocortex. In the hippocampal model of Hasselmo et al. [1], the theta rhythm alternates between two phases. In one, entorhinal input is strong and long-term potentiation is strong, but recurrent input from CA3 is weak, so new associations are encoded without interference from old ones. In the other, the balance reverses and stored associations are retrieved. That separation is implemented by oscillatory and cholinergic modulation in the hippocampus.

Cutsuridis et al. [2] showed how interneuron classes could carry out this separation. Their biophysical model of the CA1 microcircuit contains pyramidal cells and four interneuron types. Basket and axo-axonic cells, the perisomatic and axon-targeting classes that in CA1 largely express PV, fire during the storage half of the theta cycle. Bistratified and oriens lacunosum-moleculare (OLM) cells, the latter expressing SST, fire during the recall half. During recall, OLM cells inhibit the distal dendrites on which entorhinal input arrives, protecting the recalled pattern from interfering input, while bistratified cells set the threshold for recall. Removing interneuron classes one at a time tested these roles: recall survived the loss of basket and axo-axonic inhibition, but degraded under interfering entorhinal input when OLM inhibition was removed. The interneurons divide the work of the two modes between them, but the phase that selects a mode is still set from outside the circuit, by septal input pacing the theta rhythm. Whether a neocortical column could divide encoding and retrieval between its interneuron classes in a comparable way is the question this review keeps in view.

In the remainder of the chapter, Section 2.2 reviews the HTM tradition and its single model of inhibition. Section 2.3 covers associative extensions of that tradition, Sparse Distributed Memory and the Neural Association Algorithm. Section 2.4 reviews circuit models that assign distinct roles to PV, SST and VIP interneurons. Section 2.5 reviews associative-memory columns in the attractor and BCPNN tradition. Section 2.6 examines the two operator mechanisms at issue, divisive normalisation and dendritic arbitration. Section 2.7 places all of these models on the three axes.

2.2 The HTM, CLA and SDR Tradition

Hierarchical Temporal Memory (HTM) and its Cortical Learning Algorithms (CLA) form a family of models built on a small set of neocortical principles: sparse distributed representations, minicolumns, pyramidal neurons with active dendrites, and continuous local learning [3]. The tradition matters to this thesis because it provides the substrate that the thesis builds on. It is equally significant for what it leaves out.

2.2.1 Sparse Distributed Representations

A sparse distributed representation (SDR) is a large binary vector in which only a small fraction of bits are active at any time, typically a few percent. Ahmad and Hawkins [4] derived properties that make SDRs suitable for cortical computation, focusing on scaling, robustness, and generalisation. The overlap between two SDRs measures their similarity. The probability that two unrelated SDRs share many active bits falls rapidly as dimensionality grows, so a pattern can be recognised reliably by sampling only a small subset of its active bits. The same property lets a set of SDRs be stored as their union and still be matched with a low false-positive rate. Ahmad and Hawkins [5] later linked these properties to dendritic segments, showing that a segment which detects a coincidence among a small sample of active inputs behaves as a robust pattern detector over SDRs.

2.2.2 The Spatial Pooler

The Spatial Pooler (SP) converts arbitrary binary input into an SDR over a set of minicolumns [6]. Each minicolumn has a pool of potential feedforward synapses, each with a scalar permanence. A synapse counts as connected when its permanence exceeds a threshold, and the minicolumn's overlap is the number of connected synapses whose inputs are active. A minicolumn becomes active if its overlap exceeds a stimulus threshold and ranks within the top s percent of its neighbourhood, with s typically 2%. The authors describe this selection as a k-winners-take-all computation. When the neighbourhood covers the whole layer, it becomes global inhibition. Learning is Hebbian: winning minicolumns strengthen synapses from active inputs and weaken synapses from inactive ones. A slower homeostatic mechanism, boosting, raises the excitability of minicolumns that are active too rarely. Cui et al. showed that the Spatial Pooler adapts quickly to changing input statistics, becomes more robust to noise with learning, uses its minicolumns efficiently, and degrades gracefully under cell death.

Dobric et al. [7, 8] found that the same homeostatic mechanism can destabilise learned representations, pushing the Spatial Pooler into unstable, "epileptic" activity. Their remedy confines homeostatic plasticity to an initial "newborn stage" and then disables it, after which learned SDRs remain stable.

2.2.3 Temporal Memory and Active Dendrites

Hawkins and Ahmad [9] extended the HTM neuron with three integration zones. The proximal zone receives feedforward input and drives action potentials. The basal zone receives contextual input, mostly from nearby cells in the same region, and the apical zone receives feedback. Each dendritic segment acts as an independent pattern detector, so a single neuron can

recognise hundreds of distinct patterns. A match on a basal or apical segment does not fire the cell. It depolarises the cell slightly, putting it in a predictive state.

The Temporal Memory (TM) network uses this predictive state for sequence learning. All cells in a minicolumn share the feedforward receptive field selected by the Spatial Pooler. When a minicolumn becomes active, its depolarised cells fire first and inhibit their neighbours, so the input is represented by a sparse, context-specific subset of cells. If no cell in the minicolumn was predicted, all of its cells become active, which signals an unanticipated input. Because predictions are sparse, the network can hold many simultaneous predictions without confusion. Cui et al. [10] showed that Temporal Memory learns variable-order sequences, keeps multiple predictions open until disambiguating input arrives, and matches the prediction accuracy of ARIMA (autoregressive integrated moving average) models, extreme learning machines, LSTM and echo-state networks on artificial and real streaming data.

2.2.4 Continual Unsupervised Learning

Learning in this lineage is local, Hebbian and continuous. There is no global error signal and no separation between training and testing. Cui et al. [10] reported continuous online learning, robustness to sensor noise, fault tolerance, and good performance without task-specific hyperparameter tuning. Iyer et al. [11] carried the dendritic principle into deep networks for continual learning. They gated units through context-modulated dendritic segments and enforced k-winners-take-all sparsity, and the combination reduced catastrophic forgetting across sequentially learned tasks.

The same property bears on the third review axis, though less directly than it might seem. Continuous learning does not make encoding and retrieval inseparable. Temporal Memory retrieves, in the sense of predicting the next element from stored sequences, while its synapses remain plastic, and each new input is matched against stored segments as it is learned. Plasticity is applied on every cycle, and nothing in the network decides that a given input should be stored rather than only recalled, or the reverse. In the terms of Section 2.1, HTM encodes and retrieves, but does not arbitrate between the two.

2.2.5 Inhibition in the HTM Lineage

Across the Spatial Pooler, Temporal Memory and their continual-learning extensions, inhibition is a single operation that ranks the candidates and keeps the top k, applied either globally or within a neighbourhood. It has no cell identity, no target compartment and no dynamics. Its effect is set by rank rather than by strength: strict kWTA imposes an upper bound of k on the

active population and selects among candidates by their ordering, not by how strongly they are driven. Fewer than k units can be active when eligibility rules intervene, such as the stimulus threshold or a local neighbourhood with few candidates, but these limits come from the eligibility criteria, not from the inhibition itself. Inhibition also does not act separately on the apical zone, even though the HTM neuron has one, so it offers no means of weighing feedback against feedforward drive.

This is a modelling choice, not a claim about biology, and the authors' own discussions point to more structure than the algorithms use. Hawkins and Ahmad [9] propose fast-spiking basket cells as the most likely source of the rapid inhibition that TM depends on. These are the parvalbumin-expressing perisomatic cells described in Section 1.3. Cui et al. [6] note that the cortex contains many distinct classes of inhibitory neurons and suggest that classes innervating cells within a minicolumn, such as bipolar and double bouquet cells, may help define it. The lineage therefore attributes its inhibition to specific and differing cell types in biology, but implements a single homogeneous rule in computation.

This contrast with the biological circuit raises two questions for the present work. The first is whether replacing k-winners-take-all with a PV-like divisive operator changes how assembly size is set, from an upper bound imposed by rank to a recruitment that depends on input magnitude, and under what conditions it can suppress the population entirely. The second, taken up together with the first in Section 2.6, is whether adding an SST-like dendritic operator gives the network a handle on feedback that it currently lacks.

2.3 Associative Extensions

The HTM algorithms reviewed above encode inputs and learn sequences within a single region. Two lines of work take sparse binary representations further, from representation to association. Sparse Distributed Memory is an earlier and independent formalism for storing and recalling patterns in a high-dimensional binary space. The Neural Association Algorithm extends the HTM tradition itself to associations across cortical areas.

2.3.1 Sparse Distributed Memory

Kanerva's Sparse Distributed Memory (SDM) stores binary words of very high dimension, typically around a thousand bits [12]. The address space is far too large for every address to have physical storage, so the memory is built from a fixed random sample of addresses, called hard locations, each holding one counter per bit. A word is written by activating every hard location within a given Hamming radius of its address and adding the word to their counters, incrementing for ones and decrementing for zeros. A word

is read by activating the hard locations around the cue address, summing their counters bit by bit, and thresholding the sums. Because each word is spread across many locations and each location holds traces of many words, reading is tolerant to noisy cues. Reading repeatedly, with each output used as the next address, converges to the stored word when the cue lies close enough to it. Storing each item at the address of its predecessor turns the same mechanism into a sequence memory. Kanerva related the architecture to the circuitry of the cerebellum.

SDM matters to this review in three ways. First, its hard locations play a role similar to the minicolumns of the Spatial Pooler, as both map an input onto a sparse set of units chosen by similarity. They differ in how that set is chosen. The SDM addresses are fixed and random rather than learned, and the active set is every location inside a radius, so its size varies with the cue. The Spatial Pooler learns its receptive fields but bounds the active set at k by rank. SDM therefore needs no inhibition to stay sparse, and its active population grows or shrinks with the cue, a property that rank-based selection does not provide (Section 2.2.5). Second, SDM separates writing from reading cleanly, as two distinct operations. That arbitration is imposed from outside, however: the caller decides whether a cycle writes or reads, and nothing in the memory makes that decision. Third, SDM is an abstract memory and not a model of cortical tissue. It has no minicolumns, dendrites or laminae, so it says nothing about how a cortical circuit might implement either its selection rule or its write and read modes.

A note on terminology is needed here. The title of this thesis describes its model as a laminar sparse-distributed memory column. The term is used in its descriptive sense: a cortical column that stores and recalls associations over sparse distributed representations. The column is built on the HTM substrate of Section 2.2, not on Kanerva's architecture of random hard locations and counters. The two share the principle that makes such memories robust, namely storage distributed across many sparsely active units. Kanerva's model is reviewed here because it is the original statement of that principle, and because it makes explicit two properties that the HTM lineage lacks: an active set whose size varies with the input, and an explicit choice between writing and reading.

2.3.2 The Neural Association Algorithm

The Neural Association Algorithm (NAA) extends HTM from a single region to a cortical space of many areas, each running the same algorithm [13]. In each area, an encoder, usually the Spatial Pooler, produces an SDR. An association function then maps the active cells of any area onto a sparse set of active cells in a target area. NAA uses the three segment types of the HTM neuron for this. Proximal segments connect to sensory input, distal (basal) segments connect to cells of other minicolumns in the same area,

and apical segments connect to cells in other areas. Each segment is active or matching, depending on how many connected synapses it has relative to two thresholds. NAA distinguishes two kinds of context: a spatial context is an undirected association between two SDRs, such as a song and the place where it was heard, whereas a temporal context is a directed association, created when one state follows another. Learning happens only inside active minicolumns, where synapses from active cells onto active segments are strengthened and new synapses are grown, while synapses onto matching segments in inactive minicolumns are weakened. Alongside the association model, the work introduces the newborn stage of the Spatial Pooler (Section 2.2.2) and a parallel implementation of minicolumn computation based on the actor programming model.

NAA is the closest predecessor of this thesis. It shares the substrate and aims at association, which an SDR column must support before encoding and retrieval can be told apart. Its scope on the other two axes is stated openly by its author. On inhibition, the implementation "omits using explicit inhibitory cells for performance reasons" and replaces them with an inhibition algorithm [13, Sec. 5.7.3]. That algorithm comes in a global and a local version, both selecting the minicolumns with the highest overlap at a fixed density, which is the k-winners-take-all rule of Section 2.2.5. No significant difference in learning performance was found between the versions, and the thesis names the investigation of inhibition as future work, specifically how the inhibition process influences sparsity and the overall performance of learning [13, Sec. 9.5.1.7]. On encoding versus retrieval, NAA learns in every cycle, as HTM does. Recalling an input from its SDR, which the thesis calls reverse encoding, is currently handled by a classifier component that is not aligned with biology, and replacing it with learned associations is again listed as future work [13, Sec. 9.5.1.2]. Finally, only the single-area core of NAA is implemented. The apical segments that would link areas, and with them any channel for feedback, are part of the model but not yet of the implementation [13, Sec. 9.5.1.1].

Its homogeneous inhibition, which the author explicitly leaves open, is a deliberate simplification of a framework built for scale, not an oversight.

2.4 Interneuron-Resolved Circuit Models

Alongside the HTM tradition, a separate body of work models cortical inhibition at the level of interneuron classes. These models take the connectivity described in Section 1.3 as their starting point. PV cells inhibit the perisomatic region of pyramidal cells, SST cells inhibit their dendrites, and VIP cells inhibit SST cells and so release the dendrites from inhibition. The models then ask what this division of labour computes. They are reviewed here in four groups: disinhibitory gating, state-dependent responses to modula-

tion, learning and optimisation within the interneuron circuit, and large-scale laminar models.

Disinhibitory gating. Yang et al. [14] proposed that the SST and VIP motif gates input pathways on the dendrites of pyramidal cells. Their pyramidal neuron has a spiking soma and several independent dendritic compartments, reduced for network simulations to a multi-compartment rate model. SST cells keep every dendritic branch inhibited by default, and top-down control signals onto VIP cells release selected branches. If the inputs of one pathway cluster on the same branches, disinhibiting those branches gates that pathway on while the others stay closed. This selectivity survives dense and even random interneuron connectivity. A calcium-based plasticity rule regulated by dendritic inhibition produces the required clustering, and the gated circuit performs a context-dependent decision task in which a rule signal selects whether motion or colour determines the response.

Hertäg and Sprekeler [15] studied how the same motif redistributes inhibition between the soma and the dendrite. Their rate model uses a two-compartment pyramidal cell, with PV cells inhibiting the soma, SST cells inhibiting the apical dendrite, and VIP cells inhibiting only SST cells. Mutual inhibition between VIP and SST cells turns weak inputs to VIP cells into large shifts between somatic and dendritic inhibition. When that mutual inhibition is strong, the circuit becomes bistable: a transient input switches it persistently between two processing modes, one in which top-down input to the apical dendrite is integrated and one in which it is cancelled. Of the models in this section, this is the closest analogue to the arbitration that this thesis is concerned with, a circuit-level switch that decides how feedback is treated. It is framed as a mode of processing top-down input, however, not as a choice between storing and recalling a memory, and the model learns no associations.

State-dependent responses. Garcia del Molino et al. [16] built a rate model of pyramidal, PV, SST and VIP populations to explain a puzzling observation from mouse visual cortex. The response of SST cells to top-down behavioural modulation, which targets VIP cells, can change sign when the visual input changes. Their analysis traces this response reversal to specific connection strengths and shows that the effect of activating one interneuron class depends on the state of the whole network. For this thesis the point is a cautionary one. A single interneuron class cannot be assigned one fixed effect from its connectivity alone, so any abstraction of PV or SST as one operator is a simplification whose limits must be stated.

Learning and optimisation. Two models show that the differentiation itself can be learned or derived. Hertäg and Sprekeler [17] added inhibitory

plasticity to a circuit of pyramidal, PV, SST and VIP cells and showed that it learns to produce prediction-error neurons, which signal mismatches between sensory input and a prediction. Keijser and Sprekeler [18] took a normative route. They optimised the connectivity and short-term plasticity of a spiking interneuron circuit for compartment-specific feedback inhibition, and the interneurons diversified during optimisation into two classes resembling PV and SST cells. The resulting circuit acts as a decoder that inverts the nonlinear computation within the pyramidal cell. This gives computational support for treating the PV and SST split as a functional one: two classes emerge when inhibition must be specific to compartments.

Laminar models. At the largest scale, Billeh et al. [19] integrated anatomical and physiological data into models of mouse primary visual cortex spanning all layers, with about 230,000 neurons in 17 cell classes. The model exists in a biophysically detailed version and a point-neuron version with identical connectivity, and both were compared against recordings from awake mice. Moreni et al. [20] built a smaller model at the scale of a single column, drawing its connectivity from the same data. Their network of 5,000 leaky integrate-and-fire neurons has pyramidal, PV, SST and VIP populations in layers 2/3, 4, 5 and 6, and VIP cells alone in layer 1. Comparing two input regimes, they found that feedforward input to layer 4 produced net excitation of the column, whereas feedback-like input to layer 5 produced net inhibition, which the authors attribute to translaminar gain control through column-wide inhibition from layer 6 (see Section 1.4). The response to perturbing any single cell group also depended on the state of the column: strong feedforward input made the column less sensitive to every perturbation, while feedback input modulated interactions within it. How the column treats its inputs therefore depends on whether it is driven from below or from above, which bears directly on the arbitration question of this thesis. Like Billeh et al., however, the model has no synaptic plasticity and performs no memory task. Laminar models of this kind show that interneuron detail across layers can be assembled and constrained by data. Their question is how the circuit responds in different states, not what it stores or retrieves.

These models provide what the HTM tradition lacks on the inhibition axis. They assign distinct targets and dynamics to interneuron classes and analyse what each class contributes. On the other two axes, however, they offer much less. Their substrate is rate-coded or spiking populations, with no sparse distributed representations, no minicolumns and no learned assemblies, and their questions concern gating, modulation, error signalling and physiological fidelity. None of the models reviewed here stores a set of patterns and then recalls one from a partial cue.

2.5 Associative-Memory Columns

A third line of work asks directly whether networks of realistic cortical neurons can act as associative memories. It descends from Hebb's cell-assembly theory and its formal counterpart, the attractor network, in which a stored pattern is a stable state that the network falls into from a partial or noisy cue. Since the 1990s, Lansner and colleagues have tested this idea with increasingly detailed models of cortical layers 2/3, many of them built around a Bayesian-Hebbian learning rule known as BCPNN (Bayesian Confidence Propagation Neural Network) [21]. Because these models store and recall patterns, they are the only family reviewed so far that addresses the memory task directly. Their treatment of inhibition has also grown more differentiated over time.

Columns as memory units. Fransén and Lansner [22] modelled the horizontal network of layers 2/3 as a set of connected cortical columns, with the column rather than the single neuron as the functional unit. Each column contained biophysically detailed pyramidal cells and fast-spiking inhibitory interneurons representing basket cells. A memory was an assembly of columns linked by sparse, long-range excitatory connections. Connection strengths were not learned during the simulation. They were computed in advance with the Bayesian rule from the full set of patterns to be stored. Positive weights became excitatory connections between the pyramidal cells of different columns. Negative weights were routed onto the basket cells of the receiving column, so that inhibition between competing assemblies was disynaptic. With this wiring the network completed an assembly from a partial cue, resolved rivalry between overlapping assemblies, and sustained activity after the cue only when the excitability of its pyramidal cells was raised by simulated neuromodulation. The model thus showed that biophysically realistic neurons can support attractor memory. It used a single class of interneuron, and the switch between transient and sustained activity came from a neuromodulator, not from the inhibitory circuit.

Modular attractor networks. Later versions organised the same layer 2/3 network into hypercolumns, each containing a set of minicolumns [23]. Basket cells within a hypercolumn provide lateral inhibition that makes each module behave as a soft winner-take-all, typically allowing one active minicolumn at a time, and a stored pattern is a set of minicolumns distributed across hypercolumns. Regular-spiking non-pyramidal cells, possibly double bouquet cells, provided inhibition between minicolumns [24]. This organisation is strikingly close to the minicolumn and k-winners-take-all structure of the Spatial Pooler (Section 2.2.2). The difference is that the competition is implemented by a population of basket cells rather than by a sorting rule.

Differentiated interneurons in memory columns. Three studies added further interneuron classes to this architecture. Krishnamurthy, Silberberg and Lansner [24] introduced Martinotti cells, which receive strongly facilitating synapses from pyramidal cells, alongside basket cells, whose input synapses depress. When the network entered a stored attractor, basket cells fired first and then tapered off, while Martinotti cells engaged with a delay and shifted the balance of excitation and inhibition until the attractor terminated. A sensitivity analysis showed that Martinotti cells, rather than cellular adaptation or synaptic depression, dominated the attractor dwell time, and replacing them with additional basket cells did not reproduce the effect. The temporal profile of each interneuron class, not only the amount of inhibition, therefore shaped the memory dynamics. In a follow-up model, long-range recruitment of Martinotti cells produced surround suppression and promoted the saliency of non-uniform stimuli, and disinhibition through VIP cells removed that suppression [25]. Chrysanthidis, Fiebig and Lansner [26] addressed a different problem. In a spiking BCPNN model with hypercolumns and minicolumns, the learning rule produces negative as well as positive weights, which a single pyramidal cell cannot express under Dale's principle. Adding double bouquet cells, whose axons stay within a minicolumn, turned the learned negative weights into a plastic disynaptic pathway. Basket cells continued to provide non-specific feedback inhibition, and the network's learning and memory function were preserved.

Of the traditions reviewed, this one comes closest to the present thesis, on two of the three axes. It studies associative memory directly, and it has already shown that differentiated interneurons matter inside a memory network. The roles it assigns them are narrow, however. Basket cells implement competition between minicolumns, Martinotti cells control how long an attractor lasts and suppress its surround, and double bouquet cells make learned inhibition compatible with Dale's principle. Where the network does change mode, as in the sustained activity reported by Fransén and Lansner, the switch is neuromodulatory. The substrate also differs, since these are spiking networks whose memories are assemblies of minicolumns selected by soft winner-take-all competition, not learned SDRs over dendritic segments, and in the earliest models the stored patterns were wired in rather than learned. What interneuron-differentiated inhibition could add to this tradition is therefore not several classes, which it already has, but a division of encoding and retrieval between them.

2.6 Two Operator Mechanisms

The preceding sections point to two mechanisms that the HTM tradition lacks and that the interneuron literature associates with PV and SST cells: a divisive control of overall response magnitude, and a dendritic control of

how feedback arriving at the apical tuft is combined with feedforward drive. This section reviews each as a computational operator, states the evidence that links it to an interneuron class, and states where that link is contested.

2.6.1 Divisive Normalisation

Carandini and Heeger [27] proposed normalisation as a canonical neural computation. In its standard form, the response of a neuron is its driving input divided by a constant plus the summed activity of a pool of neurons,

$$R_j = \gamma \frac{D_j^n}{\sigma^n + \sum_k D_k^n}, \tag{2.1}$$

where D_j is the drive to neuron j, the sum runs over the normalisation pool, σ sets the input level at which normalisation takes effect, and γ and n set the gain and the steepness of the response. First developed to explain responses in primary visual cortex, normalisation has since been invoked across sensory modalities and for operations such as the effects of attention, the encoding of value and multisensory integration. Carandini and Heeger stress that a canonical computation need not have a single implementation: different circuits, regions or species may compute it with different components.

For this thesis the contrast with k-winners-take-all is the important point. Strict kWTA acts on the rank of inputs and imposes an upper bound on the number of active units. Normalisation acts on their magnitude: each response is scaled relative to the activity of the pool, and the ordering of inputs is preserved. Combined with a firing threshold, normalisation can therefore produce input-dependent recruitment, in which the number of active units follows the strength and distribution of the input rather than a bound set by rank. It does not, by itself, silence the population. If every drive in Equation 2.1 is multiplied by the same factor, each response approaches a fixed fraction of γ rather than zero, so a proportional increase in input does not drive the responses toward zero. Whether a normalising operator can suppress every unit depends on the size of the pool, the distribution of drive across it and the position of the threshold. In this thesis, complete suppression is therefore treated as a property of the proposed implementation to be tested, not as a general consequence of divisive normalisation. Input-dependent recruitment is the first property examined when a PV-like normalising operator replaces kWTA in an SDR column.

The attribution of division to PV cells, however, is not settled. Wilson et al. [28] activated and suppressed PV and somatostatin-expressing (SOM, here SST) cells optogenetically in mouse visual cortex and reported a clean dissociation: PV cells mainly divided responses, changing their gain while preserving stimulus selectivity, whereas SST cells mainly subtracted from them, shifting response levels and sharpening selectivity. Studies published

at the same time drew different conclusions. Lee et al. [29] found that activating PV cells sharpened orientation and direction tuning and improved orientation discrimination in behaving mice, while activating SST or VIP cells had no significant effect. Atallah et al. [30] found that PV activity transformed responses linearly, subtracting and scaling them with little change in tuning width. El-Boustani and Sur [31] then showed that the operation performed by SST cells depends on the stimulus: for small stimuli SST cells respond later than their targets and inhibit subtractively, but for large stimuli they respond at the same time and inhibit divisively. They concluded that the arithmetic of inhibition is a dynamic property of the circuit, not a fixed property of a cell class. Seybold et al. [32] reached a similar conclusion from a different direction, showing that recurrent network integration can turn subtractive inhibition of single neurons into divisive inhibition of the population, or the reverse.

This evidence fixes how the operators are used in this thesis. As stated in Section 1.3, the synaptic targeting of PV and SST cells is robust, while the assignment of a fixed arithmetic operation to each class is not. The thesis therefore treats "PV as divisive normalisation" and "SST as dendritic gating" as operator hypotheses: modelling choices motivated by targeting and by the studies above, whose consequences are tested rather than assumed. This is also why the comparison with kWTA is framed as a falsification check. If a PV-like normalising operator in an SDR column behaves no differently from kWTA, the operator distinction has no computational consequence at this level of abstraction, and the thesis reports that result.

2.6.2 Dendritic and Apical Arbitration

The second operator concerns where inhibition acts on the pyramidal cell rather than how strongly. Larkum, Zhu and Sakmann [33] showed that layer 5 pyramidal neurons have a second spike-initiation zone in the distal apical dendrite. Input to the apical tuft alone is strongly attenuated before it reaches the axon. If a back-propagating action potential from the soma coincides with distal input within a window of a few milliseconds, however, the dendrite fires a calcium spike and the neuron emits a burst. Inhibition of the distal dendrite can block the calcium spike selectively and so prevent the burst. Larkum [34] later proposed this coupling as an organising principle of cortex. Feedforward input arrives near the soma, feedback from other areas arrives at the apical tuft in layer 1, and the pyramidal cell itself acts as an associative element that signals when the two coincide. In this view the inhibitory control of the dendrite decides whether feedback is allowed to combine with feedforward drive. Murayama et al. [35] identified the interneurons responsible in vivo. Martinotti cells in deep layers, a class of SST interneuron, inhibit the apical tufts of layer 5 neurons and set the slope and threshold of their dendritic calcium response to sensory stimulation, and

disynaptic inhibition through these cells can abolish dendritic calcium spikes entirely.

Two models have turned dendritic arbitration into a learning principle. Urbanczik and Senn [36] proposed a plasticity rule in which dendritic synapses change so as to reduce the mismatch between the dendritic potential and the somatic firing of the same neuron. Sacramento et al. [37] built on this rule in a multilayer network of multi-compartment pyramidal neurons. In their model, lateral interneurons that they identify with SST cells learn to predict the top-down feedback arriving at the apical dendrite and to cancel it. Whatever feedback is not cancelled is a local prediction error, and plasticity driven by that error approximates the backpropagation algorithm without separate learning phases. Together with the disinhibitory gating of Yang et al. [14] (Section 2.4), these models establish dendrite-targeting inhibition as the natural candidate for deciding how much feedback a pyramidal cell integrates.

Both operators are well developed in their own settings, but neither has yet been carried into the memory setting this thesis addresses. Normalisation is defined independently of any circuit, and its attribution to PV cells is contested. Apical arbitration is established at the level of single cells and of circuits for gating and learning, but the questions asked of it are coincidence detection, pathway selection and credit assignment in supervised learning. In Sacramento et al., the SST-like interneurons cancel predictable feedback so that errors can drive learning. They do not decide whether the network stores a new pattern or recalls a stored one, and the model's explicit aim of avoiding separate phases is the opposite of the mode arbitration examined here.

2.7 Synthesis: Relation to Prior Work

Table 2.1 places the models reviewed in this chapter on the three axes introduced in Section 2.1. Read by rows, it shows that each tradition is strong on one or two axes and silent on the rest. Read by columns, it shows how each axis is covered.

The HTM tradition and its associative extension supply the substrate. Sparse distributed representations over minicolumns with dendritic segments, learned continuously by local rules, are what the present model is built from. Their inhibition, however, is a single homogeneous k-winners-take-all rule, which the authors of the lineage themselves describe as a simplification (Sections 2.2.5 and 2.3.2). The interneuron-resolved circuit models supply the opposite. They give PV, SST and VIP cells distinct targets, dynamics and roles, but in rate-coded or spiking populations without sparse distributed representations, and in tasks of gating, modulation, error signalling and physiological fidelity rather than memory. The attractor and BCPNN tradition

comes closest from a third direction. It studies associative memory in cortical columns and already includes several interneuron classes, but the roles it assigns them are competition, attractor dwell time, surround suppression and compliance with Dale's principle. The two operator mechanisms, divisive normalisation and dendritic arbitration, are each well established, but outside the memory setting, and the assignment of each to a single interneuron class remains contested.

These observations are drawn together into the research gap in Chapter 3.

Model	Substrate	Inhibition model	Question tested	Encoding/ retrieval control
Hippocampal precedent (Section 2.1)				
Cutsuridis et al. [2]	Biophysical spiking CA1 microcircuit	Basket, axo-axonic, bistratified, OLM	Storage and recall across theta	External (septal theta)
HTM, CLA and SDR tradition (Section 2.2)				
Spatial Pooler, Temporal Memory [6, 9]	Binary SDRs, minicolumns, dendritic segments	Homogeneous kWTA, global or local	Sparse coding, sequence learning	Absent
Iyer et al. [11]	Deep network with dendritic segments	Homogeneous kWTA	Continual multi-task learning	Absent
Associative extensions (Section 2.3)				
SDM [12]	Abstract binary memory	None; radius selection	Storage and noisy recall	External (caller)
NAA [13]	HTM-derived SDRs, multiple areas	kWTA algorithm, no inhibitory cells	Association, SP stability	Absent
Interneuron-resolved circuit models (Section 2.4)				
Yang et al. [14]	Rate, multi-compartment pyramidal cell	PV, SST, VIP	Pathway-specific gating	Absent
Hertäg and Sprekeler [15]	Rate, two-compartment pyramidal cell	PV, SST, VIP	Somato-dendritic redistribution	Circuit-internal, feedback mode only
Garcia del Molino et al. [16]	Rate populations	PV, SST, VIP	Response reversal	Absent
Hertäg and Sprekeler [17]	Rate, plastic interneurons	PV, SST, VIP	Prediction-error learning	Absent
Keijser and Sprekeler [18]	Spiking, optimised circuit	Emergent PV- and SST-like classes	Compartment-specific inhibition	Absent
Billeh et al. [19]	Spiking laminar V1, 230,000 cells	17 data-driven cell classes	Physiological fidelity	Absent
Moreni et al. [20]	Spiking laminar column	PV, SST, VIP per layer	Feedforward vs. feedback states	Absent
Associative-memory columns (Section 2.5)				
Fransén and Lansner [22]	Biophysical spiking columns	Basket cells	Completion, rivalry	Neuro-modulatory
Lundqvist et al. [23]	Spiking hyper- and minicolumns	Basket soft WTA, RSNP cells	Attractor dynamics	Absent
Krishnamurthy et al. [24, 25]	Spiking attractor network	Basket, Martinotti (VIP in 2015)	Dwell time, surround suppression	Absent
Chrysanthidis et al. [26]	Spiking BCPNN network	Basket, double bouquet	Learned inhibition, Dale's principle	Absent
Operator mechanisms (Section 2.6)				
Normalisation [27]	Circuit-independent	Divisive	Canonical computation	Absent
Sacramento et al. [37]	Rate, multi-compartment network	SST-like apical prediction	Supervised credit assignment	Absent
This thesis	**SDR laminar column**	**PV-like divisive, SST-like dendritic**	**Encoding vs. retrieval dissociation**	**External; PV/SST division tested**

Table 2.1: Relation of this thesis to prior work on the three review axes: substrate, inhibition model, and how encoding and retrieval are arbitrated.

Bibliography

[1] Michael E. Hasselmo, Clara Bodelon, and Brad P. Wyble. "A Proposed Function for Hippocampal Theta Rhythm: Separate Phases of Encoding and Retrieval Enhance Reversal of Prior Learning." In: *Neural Computation* 14.4 (2002), pp. 793–817.

[2] Vassilis Cutsuridis, Stuart Cobb, and Bruce P. Graham. "Encoding and Retrieval in a Model of the Hippocampal CA1 Microcircuit." In: *Hippocampus* 20.3 (2010), pp. 423–446. DOI: 10.1002/hipo.20661.

[3] Jeff Hawkins, Subutai Ahmad, and Donna Dubinsky. *Hierarchical Temporal Memory including HTM Cortical Learning Algorithms*. Tech. rep. Version 0.2.1. Numenta, Inc., 2011.

[4] Subutai Ahmad and Jeff Hawkins. *Properties of Sparse Distributed Representations and their Application to Hierarchical Temporal Memory*. 2015. arXiv: 1503.07469 [q-bio.NC].

[5] Subutai Ahmad and Jeff Hawkins. *How do neurons operate on sparse distributed representations? A mathematical theory of sparsity, neurons and active dendrites*. 2016. arXiv: 1601.00720 [q-bio.NC].

[6] Yuwei Cui, Subutai Ahmad, and Jeff Hawkins. "The HTM Spatial Pooler: A Neocortical Algorithm for Online Sparse Distributed Coding." In: *Frontiers in Computational Neuroscience* 11 (2017), p. 111. DOI: 10.3389/fncom.2017.00111.

[7] Damir Dobric et al. "Improved HTM Spatial Pooler with Homeostatic Plasticity Control." In: *Proceedings of the 10th International Conference on Pattern Recognition Applications and Methods (ICPRAM)*. 2021, pp. 98–106. DOI: 10.5220/0010314200980106.

[8] Damir Dobric et al. "On the Importance of the Newborn Stage When Learning Patterns with the Spatial Pooler." In: *SN Computer Science* (2022). DOI: 10.1007/s42979-022-01066-4.

[9] Jeff Hawkins and Subutai Ahmad. "Why Neurons Have Thousands of Synapses, a Theory of Sequence Memory in Neocortex." In: *Frontiers in Neural Circuits* 10 (2016), p. 23. DOI: 10.3389/fncir.2016.00023.

[10] Yuwei Cui, Subutai Ahmad, and Jeff Hawkins. "Continuous Online Sequence Learning with an Unsupervised Neural Network Model." In: *Neural Computation* 28.11 (2016), pp. 2474–2504. DOI: 10.1162/NECO_a_00893.

[11] Abhiram Iyer et al. "Avoiding Catastrophe: Active Dendrites Enable Multi-Task Learning in Dynamic Environments." In: *Frontiers in Neurorobotics* 16 (2022), p. 846219. DOI: 10.3389/fnbot.2022.846219.

[12] Pentti Kanerva. *Sparse Distributed Memory*. Cambridge, MA: MIT Press, 1988.

[13] Damir Dobric. "Investigation and Modelling of a Cortical Learning Algorithm in the Neocortex." PhD thesis. University of Plymouth, 2024. DOI: 10.24382/5155.

[14] Guangyu Robert Yang, John D. Murray, and Xiao-Jing Wang. "A dendritic disinhibitory circuit mechanism for pathway-specific gating." In: *Nature Communications* 7 (2016), p. 12815. DOI: 10.1038/ncomms12815.

[15] Loreen Hertäg and Henning Sprekeler. "Amplifying the redistribution of somato-dendritic inhibition by the interplay of three interneuron types." In: *PLoS Computational Biology* 15.5 (2019), e1006999. DOI: 10.1371/journal.pcbi.1006999.

[16] Luis Carlos Garcia del Molino et al. "Paradoxical response reversal of top-down modulation in cortical circuits with three interneuron types." In: *eLife* 6 (2017), e29742. DOI: 10.7554/eLife.29742.

[17] Loreen Hertäg and Henning Sprekeler. "Learning prediction error neurons in a canonical interneuron circuit." In: *eLife* 9 (2020), e57541. DOI: 10.7554/eLife.57541.

[18] Joram Keijser and Henning Sprekeler. "Optimizing interneuron circuits for compartment-specific feedback inhibition." In: *PLoS Computational Biology* 18.4 (2022), e1009933. DOI: 10.1371/journal.pcbi.1009933.

[19] Yazan N. Billeh et al. "Systematic Integration of Structural and Functional Data into Multi-scale Models of Mouse Primary Visual Cortex." In: *Neuron* 106.3 (2020), 388–403.e18. DOI: 10.1016/j.neuron.2020.01.040.

[20] Giulia Moreni et al. "Cell-type-specific firing patterns in a V1 cortical column model depend on feedforward and feedback-driven states." In: *PLoS Computational Biology* 21.4 (2025), e1012036. DOI: 10.1371/journal.pcbi.1012036.

[21] Anders Lansner. "Associative memory models: from the cell-assembly theory to biophysically detailed cortex simulations." In: *Trends in Neurosciences* 32 (2009), pp. 178–186. DOI: 10.1016/j.tins.2008.12.002.

[22] Erik Fransén and Anders Lansner. "A model of cortical associative memory based on a horizontal network of connected columns." In: *Network: Computation in Neural Systems* 9.2 (1998), pp. 235–264. DOI: 10.1088/0954-898X_9_2_006.

[23] Mikael Lundqvist et al. "Attractor dynamics in a modular network model of neocortex." In: *Network: Computation in Neural Systems* 17.3 (2006), pp. 253–276.

[24] Pradeep Krishnamurthy, Gilad Silberberg, and Anders Lansner. "A cortical attractor network with Martinotti cells driven by facilitating synapses." In: *PLoS ONE* 7.4 (2012), e30752. DOI: 10.1371/journal.pone.0030752.

[25] Pradeep Krishnamurthy, Gilad Silberberg, and Anders Lansner. "Long-range recruitment of Martinotti cells causes surround suppression and promotes saliency in an attractor network model." In: *Frontiers in Neural Circuits* 9 (2015), p. 60. DOI: 10.3389/fncir.2015.00060.

[26] Nikolaos Chrysanthidis, Florian Fiebig, and Anders Lansner. "Introducing double bouquet cells into a modular cortical associative memory model." In: *Journal of Computational Neuroscience* 47 (2019), pp. 223–230. DOI: 10.1007/s10827-019-00729-1.

[27] Matteo Carandini and David J. Heeger. "Normalization as a canonical neural computation." In: *Nature Reviews Neuroscience* 13.1 (2012), pp. 51–62. DOI: 10.1038/nrn3136.

[28] Nathan R. Wilson et al. "Division and subtraction by distinct cortical inhibitory networks in vivo." In: *Nature* 488.7411 (2012), pp. 343–348. DOI: 10.1038/nature11347.

[29] Seung-Hee Lee et al. "Activation of specific interneurons improves V1 feature selectivity and visual perception." In: *Nature* 488.7411 (2012), pp. 379–383. DOI: 10.1038/nature11312.

[30] Bassam V. Atallah et al. "Parvalbumin-expressing interneurons linearly transform cortical responses to visual stimuli." In: *Neuron* 73 (2012), pp. 159–170.

[31] Sami El-Boustani and Mriganka Sur. "Response-dependent dynamics of cell-specific inhibition in cortical networks in vivo." In: *Nature Communications* 5 (2014), p. 5689. DOI: 10.1038/ncomms6689.

[32] Bryan A. Seybold et al. "Inhibitory Actions Unified by Network Integration." In: *Neuron* 87.6 (2015), pp. 1181–1192.

[33] Matthew E. Larkum, J. Julius Zhu, and Bert Sakmann. "A new cellular mechanism for coupling inputs arriving at different cortical layers." In: *Nature* 398.6725 (1999), pp. 338–341. DOI: 10.1038/18686.

[34] Matthew E. Larkum. "A Cellular Mechanism for Cortical Associations: An Organizing Principle for the Cerebral Cortex." In: *Trends in Neurosciences* 36.3 (2013), pp. 141–151.

[35] Masanori Murayama et al. "Dendritic encoding of sensory stimuli controlled by deep cortical interneurons." In: *Nature* 457 (2009), pp. 1137–1141. DOI: 10.1038/nature07663.

[36] Robert Urbanczik and Walter Senn. "Learning by the dendritic prediction of somatic spiking." In: *Neuron* 81.3 (2014), pp. 521–528. DOI: 10.1016/j.neuron.2013.11.030.

[37] João Sacramento et al. "Dendritic cortical microcircuits approximate the backpropagation algorithm." In: *Advances in Neural Information Processing Systems*. Vol. 31. 2018.